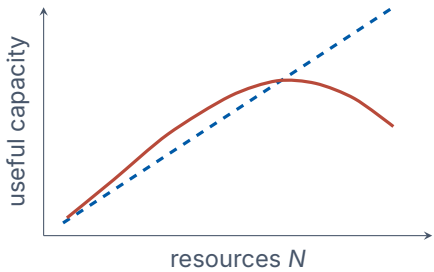


Scalable Distributed Computing - 2026/27

Lectures 5 - Sources and Limits of Scalability III

Patrizio Dazzi patrizio.dazzi@unipi.it

What are these three lessons trying to explain?

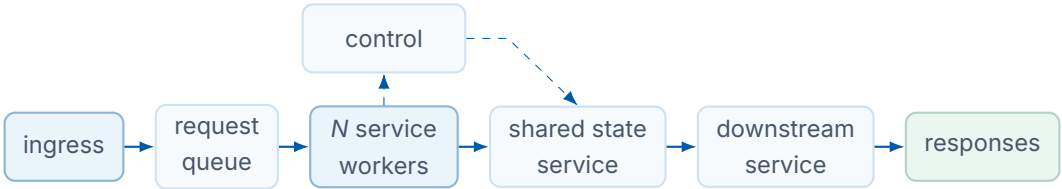


The ideal curve is easy to write. The real intellectual work begins when the measured curve departs from it.

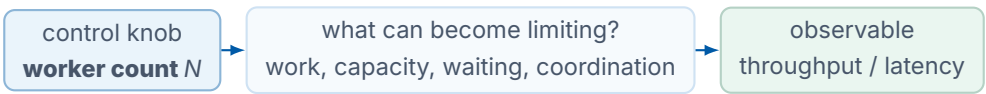
- identify a plausible mechanism;
- build the smallest useful quantitative model;
- derive a prediction that could distinguish that mechanism from alternatives.

Poor scalability is an observation, not a diagnosis.

A reference system we will keep returning to

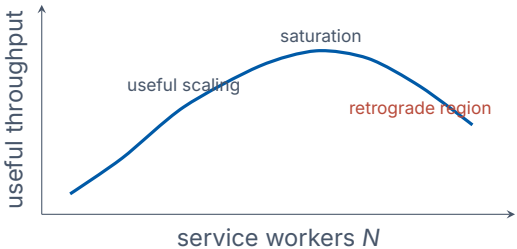


This is not one specific product. It is a deliberately small reference architecture for a common scalable service: requests enter, workers process them, every request touches shared state, and the path continues through a downstream service.



The same boxes and arrows will reappear. Each model will highlight a different reason why this end-to-end path can stop scaling.

A recurring anomaly: more workers, less useful throughput



N workers → shared state

Nothing in this path says which component is responsible. The same outward symptom can come from very different internal mechanisms.

Our task is not to name the curve. It is to explain why this system could produce it.

From interaction opportunities to a scale-dependent penalty

The previous interaction model suggests a structural growth law:

$$\text{pairwise opportunities} \propto N(N-1).$$

If a scale-dependent penalty grows approximately with this interaction structure, a simple normalized model is

$$\boxed{P_{\text{pair}}(N) = \kappa N(N-1)}.$$

Here κ is a fitted coefficient controlling the strength of the quadratic penalty.

- $N(N-1)$ captures the **growth pattern**, not a literal count of messages executed by every system;
- κ is **not** a per-message or per-pair cost;
- different interaction structures may require different growth laws.

Modelling note. The pairwise argument motivates the quadratic form. It does not constitute a causal derivation of the USL.

Build the denominator one mechanism at a time

Start with the Amdahl denominator:

$$\underbrace{1}_{\text{baseline}} + \underbrace{\sigma(N-1)}_{\text{serial / contention drag}} .$$

Now add the scale-created interaction penalty:

$$\underbrace{1}_{\text{baseline}} + \underbrace{\sigma(N-1)}_{\text{saturation pressure}} + \underbrace{\kappa N(N-1)}_{\text{coherency / cross-talk pressure}} .$$

The important modelling move is not the notation. It is separating a penalty that grows roughly linearly with additional participants from one that can grow roughly pairwise.

From the Amdahl baseline to the USL denominator

The Amdahl form gives us a first scale-dependent term whose growth is approximately linear for large N:

$$\underbrace{1}_{\text{baseline}} + \underbrace{\sigma(N-1)}_{\text{first-order penalty}} \quad \text{where} \quad \sigma(N-1) \sim \sigma N.$$

To represent a penalty that can grow much faster with scale, the USL adds a second term:

$$\boxed{D(N) = 1 + \sigma(N-1) + \kappa N(N-1)}$$

with

$$\kappa N(N-1) \sim \kappa N^2.$$

The USL denominator combines penalties with different growth rates: one roughly linear in N, the other roughly quadratic.

From the cost model to normalized capacity

The numerator N represents the ideal capacity gain from N resources. The denominator discounts that ideal gain by the two classes of scalability drag:

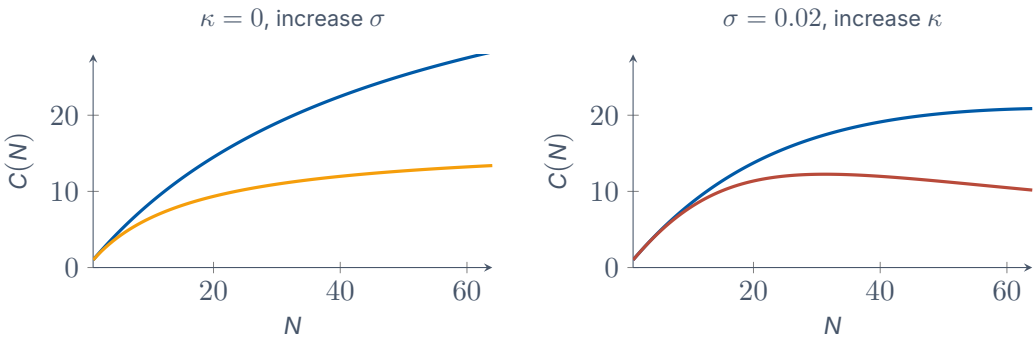
$$C(N) = \frac{N}{1 + \sigma(N-1) + \kappa N(N-1)}.$$

Read the parameters qualitatively:

- larger σ bends the curve toward an Amdahl-like plateau earlier;
- larger κ makes the scale-created penalty grow faster and can eventually bend the curve downward.

The USL is useful here because one compact expression can represent both saturation and retrograde scaling.

The two parameters bend the curve in different ways



σ mainly strengthens flattening; κ can create a peak and retrograde region.

Sanity check 1: recover ideal linear scaling

Start from

$$C(N) = \frac{N}{1 + \sigma(N-1) + \kappa N(N-1)}.$$

If both penalties vanish,

$$\sigma = 0, \qquad \kappa = 0,$$

then

$$\boxed{C(N) = N}.$$

The model contains ideal linear capacity scaling as its zero-penalty limit.

Sanity check 2: recover the Amdahl-shaped case

Set only the stronger scale-dependent term to zero:

$$\kappa = 0.$$

Then

$$C(N) = \frac{N}{1 + \sigma(N-1)},$$

which is the Amdahl-shaped normalized-capacity form.

κ is what allows the compact model to go beyond flattening and represent a true capacity peak followed by decline.

Before differentiating: build intuition for κ

The pairwise factor itself grows rapidly:

$$N(N-1).$$

For example,

$$8 \cdot 7 = 56, \qquad 64 \cdot 63 = 4032.$$

Increasing N by a factor of 8 increases this factor by a factor of 72. If $\kappa = 5 \times 10^{-4}$, its contribution to the denominator is approximately

$$0.028 \quad (N = 8), \qquad 2.016 \quad (N = 64).$$

A coefficient can be numerically tiny and still become important when it multiplies a quantity that grows rapidly with scale.

Sensitivity reveals why the quadratic term emerges at scale

Let

$$D(N) = 1 + \sigma(N-1) + \kappa N(N-1), \qquad C(N) = \frac{N}{D(N)}.$$

The local sensitivities are

$$\frac{\partial C}{\partial \sigma} = -\frac{N(N-1)}{D(N)^2}, \qquad \text{and} \qquad \frac{\partial C}{\partial \kappa} = -\frac{N^2(N-1)}{D(N)^2}.$$

Therefore

$$\boxed{\frac{\partial C/\partial \kappa}{\partial C/\partial \sigma} = N}.$$

For equal small absolute parameter changes, the local effect of the κ term grows relative to that of the σ term as N increases.

Differentiate the USL: isolate the numerator that matters

Write

$$C(N) = \frac{N}{D(N)}, \qquad D(N) = 1 + \sigma(N-1) + \kappa N(N-1).$$

Then the quotient rule gives

$$C'(N) = \frac{D(N) - ND'(N)}{D(N)^2}.$$

Since $D(N)^2 > 0$, the sign of the marginal capacity gain is controlled entirely by

$$D(N) - ND'(N).$$

Simplify the sign of the marginal capacity gain

Differentiate the denominator:

$$D'(N) = \sigma + \kappa(2N - 1).$$

Substitute into the numerator:

$$D(N) - ND'(N) = 1 - \sigma - \kappa N^2.$$

Therefore

$$\boxed{\operatorname{sign} C'(N) = \operatorname{sign}\left(1 - \sigma - \kappa N^2\right)}.$$

The capacity is still increasing while $1 - \sigma - \kappa N^2 > 0$ and decreasing once that expression becomes negative.

Solve for the continuous peak

For $\kappa > 0$ and $0 \leq \sigma < 1$, a stationary point satisfies

$$1 - \sigma - \kappa N^2 = 0.$$

Hence

$$\kappa N^2 = 1 - \sigma, \qquad \boxed{N^* = \sqrt{\frac{1-\sigma}{\kappa}}}.$$

The sign change is now explicit:

$$N < N^* \Rightarrow C'(N) > 0, \qquad N > N^* \Rightarrow C'(N) < 0.$$

The USL therefore predicts something Amdahl cannot: if $\kappa > 0$, useful capacity can reach a finite peak and then decrease.

Discrete system. N^* is a continuous approximation; real configurations require comparing nearby admissible integer values.

Small changes in κ can move the predicted useful scale substantially

Keep $\sigma = 0.03$ and use

$$N^* = \sqrt{\frac{1-\sigma}{\kappa}}.$$

Then:

κ	predicted continuous peak N^*
5×10^{-4}	≈ 44.0
10^{-3}	≈ 31.1
2×10^{-3}	≈ 22.0

Doubling κ does not merely lower the curve; because $N^* \propto \kappa^{-1/2}$, it also moves the peak left by a factor of $1/\sqrt{2}$. **Reducing scale-created interaction cost enlarges the useful operating range, not only one operating point.**

Make the peak concrete with one parameter choice

Take the curve we will plot next:

$$\sigma = 0.03, \qquad \kappa = 0.001.$$

Then

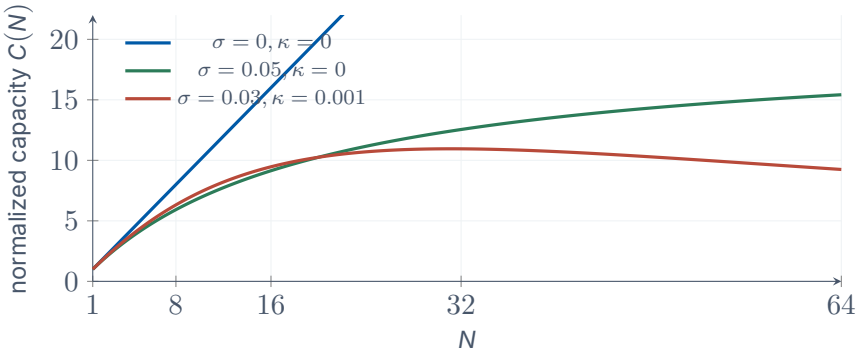
$$N^* = \sqrt{\frac{0.97}{0.001}} \approx 31.1.$$

The model predicts

$$C(16) \approx 9.47, \qquad C(31) \approx 10.95, \qquad C(64) \approx 9.25.$$

The peak is not merely a symbolic stationary point: the same model says that doubling scale from roughly 31 to 64 resources would reduce normalized capacity.

One equation, several scalability regimes



The coefficients shape the curve, but a fitted curve remains a model of observed scaling behaviour - not a root-cause report.

ACT VI

From curve to parameters

Use measured throughput to estimate the compact model.

A real dataset: ``almost linear'' can still hide measurable drag

Percona reported the following Clustrix throughput measurements:

Nodes	3	6	9	12	15	18	20
TPS	58,344	115,193	167,831	218,004	266,178	315,842	343,838

If throughput scaled exactly linearly from the 3-node measurement, 18 nodes would deliver

$$6 \times 58{,}344 = 350{,}064 \text{ TPS}.$$

The measured value is

$$315{,}842 \text{ TPS},$$

about 90% of that simple linear extrapolation.

The system looks very scalable, but the gap from ideal scaling is already large enough to measure and model.

From measurements to fitted scalability parameters



The fit tells us which compact curve is consistent with the measurements. It does **not** by itself identify the concrete subsystem responsible for the fitted penalties.

One operating point cannot tell us the shape of the scaling loss

At 18 nodes, the simple efficiency against the 3-node linear extrapolation is about

$$\frac{315{,}842}{350{,}064} \approx 0.90.$$

That single number says “about 10% below this ideal reference” - but it does not say whether the gap is:

- roughly constant as scale grows;
- bending toward a plateau;
- beginning to accelerate because of scale-created overhead.

Efficiency summarizes one point. A fitted scaling law uses the whole curve to estimate how the loss changes with scale.

Why fit a model instead of just reporting efficiency?

A single efficiency number tells us how far one operating point lies below an ideal line. A fitted scalability model tries to capture the **shape across many operating points**. That lets us ask questions such as:

- Is the loss consistent with a mainly Amdahl-like saturation term?
- Is there evidence of a scale-created term that becomes more important at larger N?
- Where would the fitted curve predict a peak if we continued scaling?

Modelling note. A better descriptive fit does not identify the physical root cause. It tells us which behavioural pattern the measurements resemble and what predictions to test next.

Why normalize throughput when $X(1)$ is known?

If baseline single-resource throughput is available, define

$$C(N) = \frac{X(N)}{X(1)}.$$

Now $C(N)$ is dimensionless:

- $C(1) = 1$ by construction;
- $C(N) = N$ would mean ideal linear scaling;
- departures from N describe scalability loss independently of the throughput unit.

Normalization separates the **shape of scaling** from the absolute speed of the system. A system can be very fast and scale poorly, or slow and scale nearly linearly.

The USL is a family of curves, not one curve

The model is

$$C(N;\sigma,\kappa) = \frac{N}{1+\sigma(N-1)+\kappa N(N-1)}.$$

The equation is known, but the parameter values are not. Different choices of

$$\boxed{(\sigma,\kappa)}$$

produce different scalability curves.

$(0.08, 0)$ and $(0.05, 0.002)$ describe two different members of the same family.

Fitting means finding which member of the model family best matches the measured points.

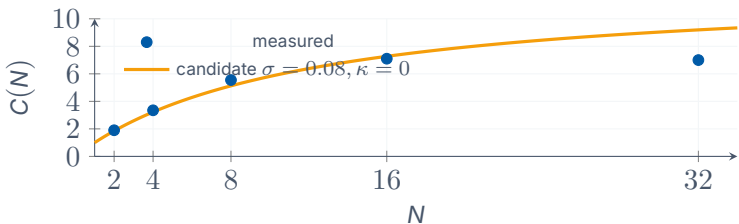
A constructed benchmark: try one candidate curve

For illustration, suppose a benchmark produced these normalized capacities:

N	2	4	8	16	32
$C_{\text{meas}}(N)$	1.90	3.35	5.55	7.10	7.00

Now try the candidate

$$\sigma = 0.08, \qquad \kappa = 0.$$



A candidate curve can be wrong by a measurable amount

At $N = 32$, the experiment measured

$$C_{\text{meas}}(32) = 7.00.$$

The candidate curve predicts

$$C_{\text{pred}}(32) = \frac{32}{1 + 0.08(31)} \approx 9.20.$$

The residual is the vertical prediction error:

$$\boxed{e_{32} = C_{\text{meas}}(32) - C_{\text{pred}}(32) \approx -2.20}.$$

One measured operating point gives one residual. A curve gives one residual at every measured operating point.

How do we judge the whole curve?

For measured points $N_1, \ldots, N_m$, define

$$e_i = C_{\text{meas}}(N_i) - C(N_i; \sigma, \kappa).$$

For the first candidate, the residuals are approximately

$$0.05, \quad 0.12, \quad 0.42, \quad -0.17, \quad -2.20.$$

Simply adding them is unsafe: positive and negative errors can cancel. A standard summary is the sum of squared residuals:

$$\boxed{E(\sigma, \kappa) = \sum_{i=1}^{m} e_i^2}$$

so that, equivalently,

$$E(\sigma, \kappa) = \sum_{i=1}^{m} \left[C_{\text{meas}}(N_i) - C(N_i; \sigma, \kappa) \right]^2 .$$

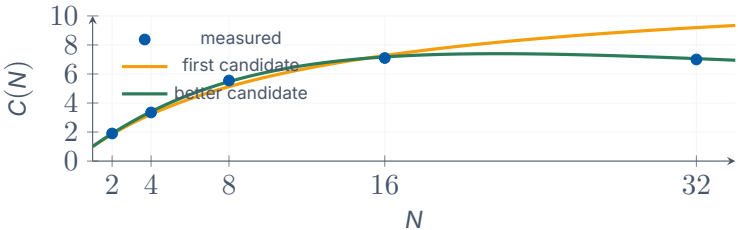
A better parameter pair gives a much better curve

Try instead

$$\sigma = 0.05, \qquad \kappa = 0.002.$$

For the same measurements,

$$E(0.08, 0) \approx 5.05, \qquad \boxed{E(0.05, 0.002) \approx 0.018}.$$



Fitting is an optimization problem over parameter values

Every pair (σ, κ) produces a curve, and every curve produces an overall error

$$E(\sigma, \kappa).$$

The fitted parameters are the values that minimize that discrepancy:

$$\boxed{(\hat{\sigma}, \hat{\kappa}) = \arg\min_{\sigma,\kappa} E(\sigma, \kappa)}.$$

In practice, numerical software performs the search.

We are fitting the **nonlinear USL curve**. The optimizer changes the parameters; it does not turn the performance curve into a straight line.

For real data, the objective may also account for measurement uncertainty, weighting, or parameter constraints.

Why do we care about the fitted parameters?

For the illustrative fit

$$\sigma = 0.05, \qquad \kappa = 0.002,$$

the continuous USL peak is

$$N^* = \sqrt{\frac{1-\sigma}{\kappa}} = \sqrt{\frac{0.95}{0.002}} \approx 21.8.$$

This does **not** mean "deploy exactly 22 nodes." It gives an operational hypothesis:

- configurations around this region deserve denser measurement;
- scaling far beyond it may no longer increase capacity under the fitted regime;
- a redesign that reduces the fitted scale-dependent penalty should move the useful range outward.

After fitting: what does the fitted penalty do as N grows?

The nonlinear fit gave us values for σ and κ in

$$C(N) = \frac{N}{1 + \sigma(N-1) + \kappa N(N-1)}.$$

That answers one question:

Which USL curve best matches the measured capacities?

We now ask a different one:

> How does the loss represented by that fitted curve grow as the system gets larger?

In particular, we want to distinguish a loss that grows only because there are more participants from a loss whose **intensity also increases with scale**.

Why this distinction matters

Consider two USL curves with the same $\sigma = 0.05$ but different κ:

$$\kappa = 0 \qquad \text{and} \qquad \kappa = 0.002.$$

N	$C(N)$, $\kappa = 0$	$C(N)$, $\kappa = 0.002$
4	3.48	3.41
16	9.14	7.17
32	12.55	7.06

At small scale the two curves are close. At larger scale their behaviour is very different.

A scale-dependent penalty may be almost invisible in a small experiment and dominate later.

Extract the loss implied by a measured capacity

The USL has the generic form

$$C(N) = \frac{N}{1 + \text{penalty}}.$$

So, from a measured or fitted capacity value $C(N)$, the model-implied penalty is

$$\boxed{P(N) = \frac{N}{C(N)} - 1}.$$

This definition has the expected reference behaviour:

$$C(N) = N \quad \Rightarrow \quad P(N) = 0.$$

If capacity falls below ideal linear scaling, then

$$C(N) < N \quad \Rightarrow \quad P(N) > 0.$$

Modelling note. $P(N)$ is dimensionless and model-implied. It is not directly an elapsed time, message count, or lock-wait measurement.

Example: the total implied penalty grows rapidly

Use the illustrative fitted parameters

$$\sigma = 0.05, \qquad \kappa = 0.002.$$

For three operating points:

N	fitted $C(N)$	$P(N) = N/C(N) - 1$
4	3.41	0.174
16	7.17	1.230
32	7.06	3.534

The total model-implied penalty increases by more than a factor of 20 between $N = 4$ and $N = 32$.

The next question is why it grows: simply because there are more participants, or because the penalty associated with scale is itself becoming stronger?

The USL shows the two growth contributions explicitly

For the USL,

$$P(N) = \sigma(N-1) + \kappa N(N-1).$$

The two terms have different growth laws:

$$\underbrace{\sigma(N-1)}_{\text{approximately linear in } N} + \underbrace{\kappa N(N-1)}_{\text{approximately quadratic in } N}.$$

For large N,

$$\sigma(N-1) \sim \sigma N, \qquad \kappa N(N-1) \sim \kappa N^2.$$

The fitted model itself contains a term whose contribution can grow much faster than the number of resources.

Total penalty mixes system size and penalty intensity

Rewrite the same expression as

$$P(N) = (N-1)(\sigma + \kappa N).$$

This separates two factors:

$$\underbrace{N-1}_{\text{how many participants beyond the baseline}} \times \underbrace{(\sigma + \kappa N)}_{\text{remaining scale-dependent penalty factor}}.$$

So an increasing $P(N)$ can come from two effects at once:

- there are more participants;
- the penalty factor associated with the operating scale may itself be increasing.

To study the second effect, remove the simple multiplicity factor $N-1$.

Normalize out the number of additional participants

For $N > 1$, define

$$\boxed{Q(N) = \frac{P(N)}{N-1}}.$$

Substituting the USL expression for $P(N)$ gives

$$Q(N) = \frac{(N-1)(\sigma + \kappa N)}{N-1} = \boxed{\sigma + \kappa N}.$$

$Q(N)$ therefore answers a more focused question:

After accounting for the number of additional participants, how does the remaining penalty factor change with scale?

The transformed quantity has a direct model interpretation

The USL predicts

$$Q(N) = \sigma + \kappa N.$$

Two cases are immediately different:

$$\kappa = 0 \quad \Rightarrow \quad Q(N) = \sigma \quad \text{(constant with scale)},$$

while

$$\kappa > 0 \quad \Rightarrow \quad Q(N) \text{ increases with } N.$$

For one additional unit of scale,

$$Q(N+1) - Q(N) = \kappa.$$

Modelling note. κ is the slope of this normalized model penalty. It is not a literal per-message or per-pair cost.

Numerical example: the normalized penalty increases with scale

Continue with

$$\sigma = 0.05, \qquad \kappa = 0.002.$$

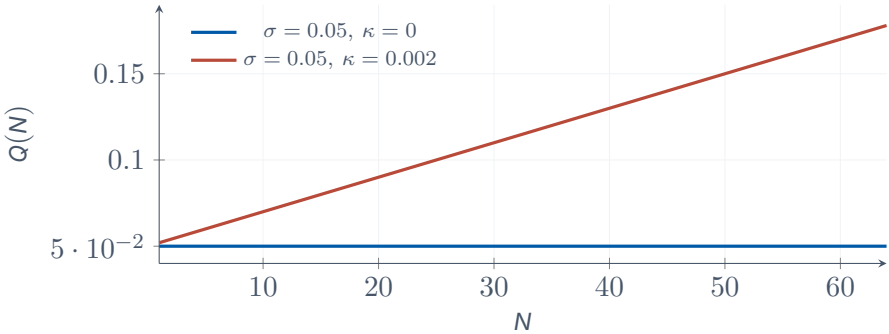
N	$P(N)$	$Q(N) = P(N)/(N-1)$	$\sigma + \kappa N$
4	0.174	0.058	0.058
16	1.230	0.082	0.082
32	3.534	0.114	0.114

The total penalty $P(N)$ grows very quickly, but even after removing the factor $N - 1$, the remaining quantity still rises:

$$0.058 \longrightarrow 0.082 \longrightarrow 0.114.$$

The loss is not increasing only because there are more participants; the normalized penalty factor itself is increasing with scale.

Compare a constant penalty factor with an increasing one



With $\kappa = 0$, the normalized penalty factor stays flat. With $\kappa > 0$, the model assigns a larger penalty factor at larger operating scales.

Return to the total penalty: this is where the quadratic term comes from

We have

$$P(N) = (N-1)Q(N)$$

and

$$Q(N) = \sigma + \kappa N.$$

Therefore

$$P(N) = (N-1)(\sigma + \kappa N) = \sigma(N-1) + \kappa N(N-1).$$

For large N, the κ contribution behaves as

$$\boxed{\kappa N(N-1) \sim \kappa N^2}.$$

So the quadratic growth has a simple two-factor interpretation:

$$\underbrace{N}_{\text{number of participants}} \times \underbrace{N}_{\text{penalty factor increasing with scale}} \Rightarrow N^2.$$

This reconnects with the earlier interaction-growth intuition

Earlier we used potential pairwise interactions as a structural motif:

$$\text{interaction opportunities} \propto N(N-1).$$

The fitted USL contains a penalty contribution with the same growth law:

$$\text{USL scale-dependent contribution} = \kappa N(N-1).$$

These two statements are compatible:

$$N(N-1) \sim N^2.$$

But they are not the same claim.

Modelling note. Observing a quadratic fitted penalty does not prove that the real system performs all-to-all communication. Different mechanisms can generate similar growth laws.

The straight line is not the performance curve

The original quantity of interest is normalized capacity:

$$C(N) = \frac{N}{1 + \sigma(N-1) + \kappa N(N-1)}$$

which is nonlinear in N.

The transformed quantity is

$$Q(N) = \frac{N/C(N) - 1}{N - 1} = \sigma + \kappa N$$

which is linear in N under the USL.

Original coordinates	Transformed coordinates
plot $C(N)$ against N	plot $Q(N)$ against N
nonlinear capacity curve	straight penalty-factor line
answers “how much capacity?”	answers “how does the normalized penalty grow?”

What the transformed view is useful for

Once a nonlinear USL fit has been obtained, the transformed quantity

$$Q(N) = \frac{N/C(N) - 1}{N - 1}$$

provides a compact diagnostic view.
It lets us inspect whether, over the measured range,

- the normalized penalty factor is approximately constant;
- it increases systematically with scale;
- the transformed measurements are broadly compatible with the USL prediction $Q(N) = \sigma + \kappa N$.

The intercept and slope have an immediate model interpretation:

$$\boxed{\text{intercept} = \sigma} \qquad \boxed{\text{slope} = \kappa}.$$

If the one-resource baseline is unavailable

The transformed quantity requires normalized capacity

$$C(N) = \frac{X(N)}{X(1)}.$$

If $X(1)$ was not measured, do not invent it. Fit throughput directly:

$$\boxed{X(N) = \lambda \frac{N}{1 + \sigma(N-1) + \kappa N(N-1)}}$$

where, in the model,

$$X(1) = \lambda.$$

The same nonlinear fitting principle applies; the optimizer now estimates λ, σ, and κ together.

Next research question: does the model describe a real benchmark?

The constructed example showed how fitting and transformation work when the data are generated from a known USL curve.
With real measurements, three questions must be separated:

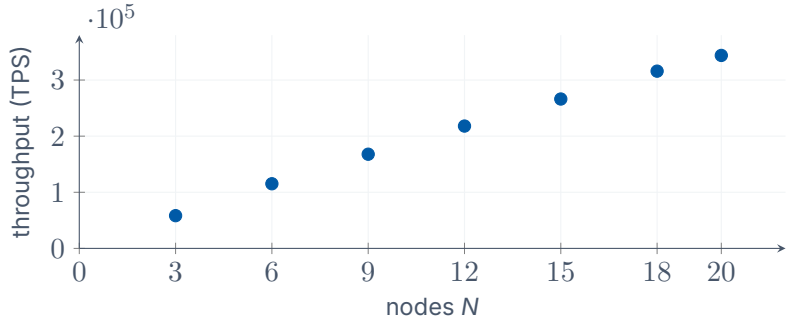
1. **descriptive fit:** does the USL curve follow the measured points over the observed range?
2. **extrapolation:** what does the fitted curve predict outside that range?
3. **causality:** do the fitted parameters identify the physical mechanism producing the loss?

The next slides address these questions in that order.

Real benchmark data before fitting

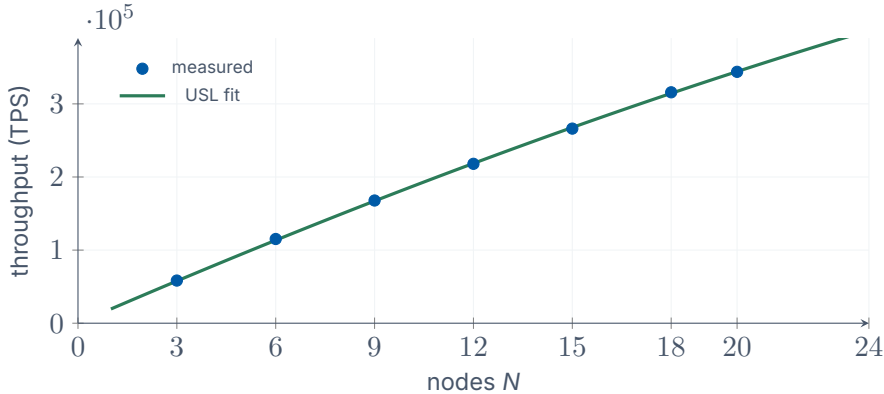
Percona reported the following Clustrix throughput measurements:

Nodes	3	6	9	12	15	18	20
TPS	58,344	115,193	167,831	218,004	266,178	315,842	343,838



Schwartz, Percona Live 2011; Clustrix dataset/refit.

Add the fitted USL curve only after inspecting the data



Clustrix throughput data and USL refit reported by Percona.

The reported fit estimates three quantities

The reported refit used an estimated

$$X(1) = 19{,}448 \text{ TPS}$$

and obtained

$$\boxed{\sigma \approx 0.00509}, \qquad \boxed{\kappa \approx 8.79 \times 10^{-5}}.$$

The reported goodness of fit was

$$R^2 \approx 0.99998.$$

These numbers describe the relationship between the USL model and this dataset over the measured range.

What does the real-data fit establish?

Over the measured range $3 \leq N \leq 20$:

- the fitted USL curve is very close to the reported throughput points;
- the fitted σ and κ are numerically small;
- the dataset is therefore consistent with a nearly linear scaling regime over the observed node counts.

This supports the USL as a compact **descriptive model** for these measurements.

Modelling note. Good descriptive agreement over the observed range is not evidence that the model has identified a unique physical mechanism.

The fitted peak lies outside the measured range

For the reported parameters,

$$N^* = \sqrt{\frac{1-\sigma}{\kappa}} \approx 106.$$

But the largest measured configuration was

$$N = 20.$$

So the statement

$$N^* \approx 106$$

is an extrapolation of the fitted model by more than a factor of five in resource count.

The data establish the measured points up to $N = 20$; the peak near 106 is a model prediction conditional on the fitted regime continuing.

A fitted curve does not uniquely identify the physical cause

The USL fit provides a compact description of the observed capacity shape.
However, several different system mechanisms can produce reduced marginal gain or eventual decline, including

- a fixed shared bottleneck;
- contention and queueing around shared resources;
- increasing communication or coordination work;
- synchronization and slowest-participant effects;
- load imbalance.

Therefore the fitted values of σ and κ are not, by themselves, a root-cause diagnosis.

Diagnosis requires predictions beyond the fitted curve -- I

For each candidate explanation, ask what else should be observable if that explanation is correct.

Hypothesis	Additional observable consequence
Serial work	a persistent non-accelerated path remains on the critical path
Fixed bottleneck	utilization or saturation concentrates at the limiting shared resource
Contention	queueing, blocking, or lock-wait time grows as N increases

The curve suggests where to investigate; these additional observations test the explanation.

Diagnosis requires predictions beyond the fitted curve -- II

Other hypotheses imply different observations:

Hypothesis	Additional observable consequence
Communication overhead	communication or data-movement work grows relative to useful work
Synchronization	completed participants remain idle while the slowest participant finishes
Imbalance	$\max_i W_i$ grows relative to the average participant work
Coordination / coherency	coordination work per useful operation grows with scale

Different hypotheses therefore lead to different measurements and interventions.

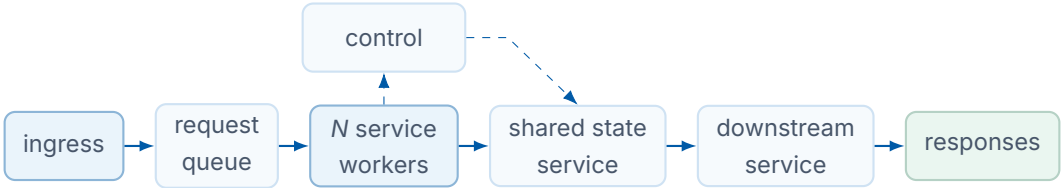
A useful diagnosis changes what experiment we run next

Suppose throughput falls after $N = 32$. Three hypotheses suggest three different interventions:

Hypothesis	Intervention	Expected response
fixed bottleneck	raise capacity of the suspected shared tier	plateau / peak should move right
contention	reduce parallel entrants to the shared path	waiting should fall; throughput may recover
scale-created communication	reduce fan-out / coordination work per operation	high-N degradation should weaken

The strongest explanation is not the one with the nicest curve fit; it is the one whose additional predictions survive targeted intervention.

Back to the reference system

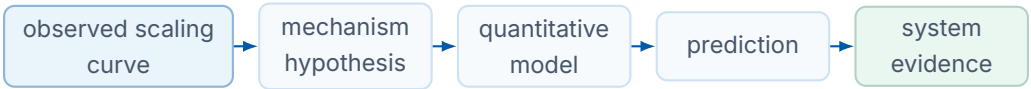


A throughput curve that bends or declines does not tell us which internal component or interaction is responsible. Our candidate explanations now make different predictions:

non-scalable critical path	shared-state capacity	waiting / contention	control / coordination work

The curve is the symptom. The model tells us what additional evidence would support or reject a mechanism.

The unifying view



Scalability is not a curve to admire. It is a phenomenon to explain.

The next lecture on workloads and measurement supplies the methodology needed to test these predictions correctly.

References used in this lecture

- G. M. Amdahl, “Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities,” AFIPS, 1967.
- J. Dean and S. Ghemawat, “MapReduce: Simplified Data Processing on Large Clusters,” OSDI, 2004.
- N. J. Gunther, “A General Theory of Computational Scalability Based on Rational Functions,” 2008; and *Guerrilla Capacity Planning*.
- B. Schwartz, “Forecasting MySQL Scalability with the Universal Scalability Law,” Percona Live, 2011.
- A. B. Bondi, “Characteristics of Scalability and Their Impact on Performance,” WOSP, 2000. (background framing from earlier in the module)

Empirical examples are explicitly separated from guided numerical examples, reference-system diagrams, and deliberately simplified analytical models.



Questions or comments?

L5 - Sources and limits of scalability III

Department of Computer Science - University of Pisa